Program: 1. Demonstrate Node .Js Application to perform CRUD operation for online Book Cart

Prelims which is needed before running the programs:

```
$npm init -y
$npm install express
```

Example Program:

```
const express = require('express');
const app = express();
const port = 3000;


// Middleware to parse JSON bodies
app.use(express.json());


// Our in-memory JSON data store
let books = [
  { id: 1, title: 'The Lord of the Rings', author: 'J.R.R.
Tolkien', price: 25.00, quantity: 10 },
  { id: 2, title: 'Pride and Prejudice', author: 'Jane Austen', price: 15.50, quantity: 5 },
  { id: 3, title: 'To Kill a Mockingbird', author: 'Harper Lee', price: 10.00, quantity: 7 }
];


app.get('/books', (req, res) => {
  res.status(200).json(books);
});

app.get('/books/:id', (req, res) => {
  const book = books.find((b) => b.id === parseInt(req.params.id));
  console.log(typeof(req.params.id)," ",book);
  if(book){
    res.status(200).json(book);
  }else{
    res.status(201).send(" book not found");
  }

});









app.post('/books', (req, res) => {
  const newBook = {
```

```
    id : books.length + 1,
    title:req.body.title,
    author:req.body.author,
    price:req.body.price,
    quantity:req.body.quantity
  };
  books.push(newBook);
  res.status(201).json(newBook);
});



app.patch('/books',(req,res,err)=>{
  res.send("in the patch method");
});



app.put('/books',(req,res,err)=>{
  res.send("in the put method");
});


app.delete('/books/:id', (req, res) => {
  const initialLength = books.length;
  books = books.filter(book => book.id !== parseInt(req.params.id));

  if (books.length === initialLength) {
    return res.status(404).json({ message: 'Book not found' });
  }

  res.status(200).json({ message: 'Book deleted successfully' });
});




app.listen(port, () => {
  console.log(`Server is running on http://localhost:${port}`);
});
```

Expected OUTPUT:

Program 2: Write a node.js program using Express framework to accept user name, Branch, Semester, from web page and display the information as below
a) Handle both get and post methods
b) Branch should be underlined
c) Name should be in bold face.

Prelims which is needed before running the programs:

```
$npm init -y
$npm install express
```

Program: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Student Information</title>
   <style>
      body { font-family: Arial, sans-serif; padding: 20px; }
      .container { max-width: 500px; margin: auto; }
      h2 { text-align: center; }
      form { margin-bottom: 20px; border: 1px solid #ccc; padding: 20px; border-
radius: 8px; }
      label { display: block; margin-bottom: 5px; }
      input[type="text"] { width: 100%; padding: 8px; margin-bottom: 10px; box-
sizing: border-box; }
      button { padding: 10px 15px; cursor: pointer; }
      .output { margin-top: 20px; padding: 15px; border: 1px solid #007BFF;
background-color: #e6f2ff; border-radius: 8px; }
   </style>
</head>
<body>
   <div class="container">
      <h2>Student Information Form</h2>

      <form action="/submit-get" method="get">
         <h3>Using GET Method</h3>
         <label for="name-get">Name:</label>
         <input type="text" id="name-get" name="name" required>
         <label for="branch-get">Branch:</label>
         <input type="text" id="branch-get" name="branch" required>
         <label for="semester-get">Semester:</label>
         <input type="text" id="semester-get" name="semester" required>
         <button type="submit">Submit (GET)</button>
      </form>



      <form action="/submit-post" method="post">
```

```
        <h3>Using POST Method</h3>
        <label for="name-post">Name:</label>
        <input type="text" id="name-post" name="name" required>
        <label for="branch-post">Branch:</label>
        <input type="text" id="branch-post" name="branch" required>
        <label for="semester-post">Semester:</label>
        <input type="text" id="semester-post" name="semester" required>
        <button type="submit">Submit (POST)</button>
      </form>
    </div>
</body>
</html>
```

Example Program: server.js

```
const express = require('express');
const path = require('path');
const app = express();
const port = 3000;

// Middleware to serve static files (like your HTML file)
app.use(express.static(path.join(__dirname)));

// Middleware to parse URL-encoded data (for POST requests)
app.use(express.urlencoded({ extended: true }));

// GET method handler
app.get('/submit-get', (req, res) => {
    // Data is in req.query for GET requests
    const name = req.query.name;
    const branch = req.query.branch;
    const semester = req.query.semester;

    // Display the information with specified formatting
    const htmlResponse = ` <h2>Student Information (GET)</h2>
        <p>Name: <b>${name}</b></p>
        <p>Branch: <u>${branch}</u></p>
        <p>Semester: ${semester}</p>
        <br> <a href="/">Go Back</a>`;
    res.send(htmlResponse);
});
```

```
// POST method handler
app.post('/submit-post', (req, res) => {
```

```
    // Data is in req.body for POST requests
    const name = req.body.name;
    const branch = req.body.branch;
    const semester = req.body.semester;

    // Display the information with specified formatting
    const htmlResponse = `
        <h2>Student Information (POST)</h2>
        <p>Name: <b>${name}</b></p>
        <p>Branch: <u>${branch}</u></p>
        <p>Semester: ${semester}</p>
        <br>
        <a href="/">Go Back</a>
    `;
    res.send(htmlResponse);
});

// Start the server
app.listen(port, () => {
    console.log(`Server is listening at http://localhost:${port}`);
    console.log('Open your browser and navigate to
http://localhost:3000/index.html');
});
```

$node server

go to this URL --> http://localhost:3000/index.html

Expected OUTPUT

Design a resume of a job aspirant using React components like Classes and Functions. Style the resume by applying CSS

Prelims which is needed before running the programs:

$npx create-react-app program3

Example Program: /src/ResumeCreate/Education.js

```
import React from "react";

class EdTable extends React.Component {
    render() {
        const education = [
            {
                id: 1,
                name: "National Public School",
                place: "Bengaluru",
                stddeg: "10th",
                grade: "A+",
            },
            {
                id: 2,
                name: "Indian Institute of Science",
                place: "Bengaluru",
                stddeg: "under grade",
                grade: "9.8",
            },
        ]
const edudetails = education.map(educ =>
                <EdRow id={educ.id}
                        name={educ.name}
                        place={educ.place}
                        stddeg={educ.stddeg}
                        grade={educ.grade} /> );
        return (
            <table>
                <tr>
                    <th>id</th>
                    <th>name</th>
                    <th>place</th>
                    <th>education</th>
                    <th>grade</th>
                </tr>
                <tbody>
                    {edudetails}
                </tbody>
            </table>
        );
    }
}
class EdRow extends React.Component {
    render() {
        return (
            <tr>
                <td>{this.props.id}</td>
                <td>{this.props.name}</td>
```

```
                <td>{this.props.place}</td>
                <td>{this.props.stddeg}</td>
                <td>{this.props.grade}</td>
            </tr>
        );
    }
}
class Exducation extends React.Component {
    render() {
        return (
            <section>
                <EdTable />
            </section>
        );
    }
}

export default Education;
```

Example Program: Resume_Header.js

```
import React from "react";

class Resume_Header extends React.Component{
  render(){
    return(
      <h1>Resume Details</h1>
    );
  }
}

export default Resume_Header;
```

Example Program: Resume_Person_Info.js

```
import React from "react";
class Resume_Person_Info extends React.Component {
  render() {
    return (
      <div >
        <div > Name : <p>Prashanth K</p></div><br/>
         <div >Address : <p>Kengeri</p></div><br/>
         <div >Phone No : <p>1234567890</p></div><br/>
         <div >email id : <p>prashanthk@rvce.edu.in</p></div>
      </div>
    );
  }
}
export default Resume_Person_Info;
```

Example Program: Index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
```

```
import Resume_Header from './ResumeCreate/Resume_Header';
import Resume_Person_Info from './ResumeCreate/Resume_Person_Info';
import Education from './ResumeCreate/Educational';

const root = ReactDOM.createRoot(document.getElementById('root'));
    root.render(
  <React.StrictMode>
   <Resume_Header />
    <Resume_Person_Info />
    <Education />
  </React.StrictMode>
);
```

Steps for execution:

$npm start

> **prog3@0.1.0 start**
>
> **react-scripts start**

Expected OUTPUT

Program 4: Build student registration portal using Entities like component, state and props

Prelims which is needed before running the programs:

$npx create-react-app program4

Example Program: /src/Student.js

```
import React from "react";

const studlist = [{
   slno: 1,
   name: "stud1",
   usn: "rvce1",
   tmarks: 150,
}, {
   slno: 2,
   name: "stud2",
   usn: "rvce2",
   tmarks: 145,
}]

class StudentRow extends React.Component {
   render() {
      const studl = this.props.stdli;
      return (
         <tr>
            <th>{studl.slno}</th>
            <th>{studl.name}</th>
            <th>{studl.usn}</th>
            <th>{studl.tmarks}</th>
         </tr>
      );
   }
}
```

```
class StudentsTable extends React.Component {
   render() {
      const studrows = this.props.studlist.map(studli =>
         <StudentRow key={studli.slno} stdli={studli} />
      );
      return (
```

```
            <table>
              <tr>
                <th>Sl No</th>
                <th>Student Name</th>
                <th>USN</th>
                <th>Total Marks</th>
              </tr>
              <tbody>
                {studrows}
              </tbody>
            </table>
        );
    }
}

class StudentAddSubmit extends React.Component {
    constructor() {
        super();
        this.handleSubmit = this.handleSubmit.bind(this);
    }
    handleSubmit(e) {
        e.preventDefault();
        const form = document.forms.studAdd;
        const studli = {
            name: form.name.value,
            usn: form.usn.value,
            tmarks: form.tmarks.value,
        }
        console.log("studli - ",studli);
        this.props.CreateStudentAdd(studli);
        form.name.value = ""; form.usn.value = ""; form.tmarks.value = ""
    }

    render() {
        return (
            <form name="studAdd" onSubmit={this.handleSubmit}>
                <input type="text" name="name" placeholder="Name" />
                <input type="text" name="usn" placeholder="USN" />
                <input type="text" name="tmarks" placeholder="Total Marks" />
                <button>Add</button>
            </form>
        );
    }
}



class StudentListTable extends React.Component{
    constructor() {
        super();
        this.state = { newStudList: [] };
        this.CreateStudentAdd = this.CreateStudentAdd.bind(this);
    }
```

```
  componentDidMount() {
    this.loadData();
  }

  loadData() {
    setTimeout(() => {
      this.setState({ newStudList: studlist })
    }, 5000);
    console.log("i have loaded the student add component also");
  }

  CreateStudentAdd(studli) {
    const studlength = this.state.newStudList.length+1;
    studli.slno = studlength;
    const newstudlist1 = this.state.newStudList.slice();
    newstudlist1.push(studli);
    this.setState({ newStudList: newstudlist1 });
  }
  render(){
    return(
          <React.Fragment>
            <hr/>
            <StudentsTable studlist = {this.state.newStudList}/>
            <hr/>
            <StudentAddSubmit CreateStudentAdd={this.CreateStudentAdd}/>
          </React.Fragment>
    );
  }
}


export default StudentListTable;
```

Steps for execution:

$npm start

> prog3@0.1.0 start

> react-scripts start

Expected OUTPUT

Design and implement a React Form that collects user input for name, email, and password. Validate the form using Regular Expression.

Prelims which is needed before running the programs:

$npx create-react-app program5

Example Program: /src/RegistrationForm.js

```
import React from "react";

class RegistrationForm extends React.Component {
  state = {
    name: "",
    email: "",
    password: "",
    errors: {}
  };

  validate() {
    const errors = {};
    if (!this.state.name) {
      errors.name = "Name is required";
    } else if (this.state.name.length < 3) {
      errors.name = "Name must be at least 3 letters";
    }
    if (!this.state.email) {
      errors.email = "Email is required";
    } else if (!/\S+@\S+\.\S+/.test(this.state.email)) {
      errors.email = "Email is invalid";
    }
    if (!this.state.password) {
      errors.password = "Password is required";
    } else if (this.state.password.length < 8) {
      errors.password = "Password must be at least 8 letters";
    }
    this.setState({ errors });
    return Object.keys(errors).length === 0;
  };

  handleChange(e){
    this.setState({ [e.target.name]: e.target.value });
  };

  handleSubmit(e){
    e.preventDefault();
    if (this.validate()) {
      alert("Form submitted!");
      // Reset form or further action
      this.setState({ name: "", email: "", password: "", errors: {} });
    }
  };

  render() {
    const { name, email, password, errors } = this.state;

    return (
      <div><h1>Registration form for USER Input:</h1>
      <form onSubmit={this.handleSubmit}>
        <input
          name="name"
          placeholder="Name"
          value={name}
```

```
          onChange={this.handleChange}
        />
        <div>{errors.name}</div>

        <input
          name="email"
          placeholder="Email"
          value={email}
          onChange={this.handleChange}
        />
        <div>{errors.email}</div>

        <input
          name="password"
          type="password"
          placeholder="Password"
          value={password}
          onChange={this.handleChange}
        />
        <div>{errors.password}</div>

        <button type="submit">Register</button>
      </form>
      </div>
    );
  }
}

export default RegistrationForm;
```

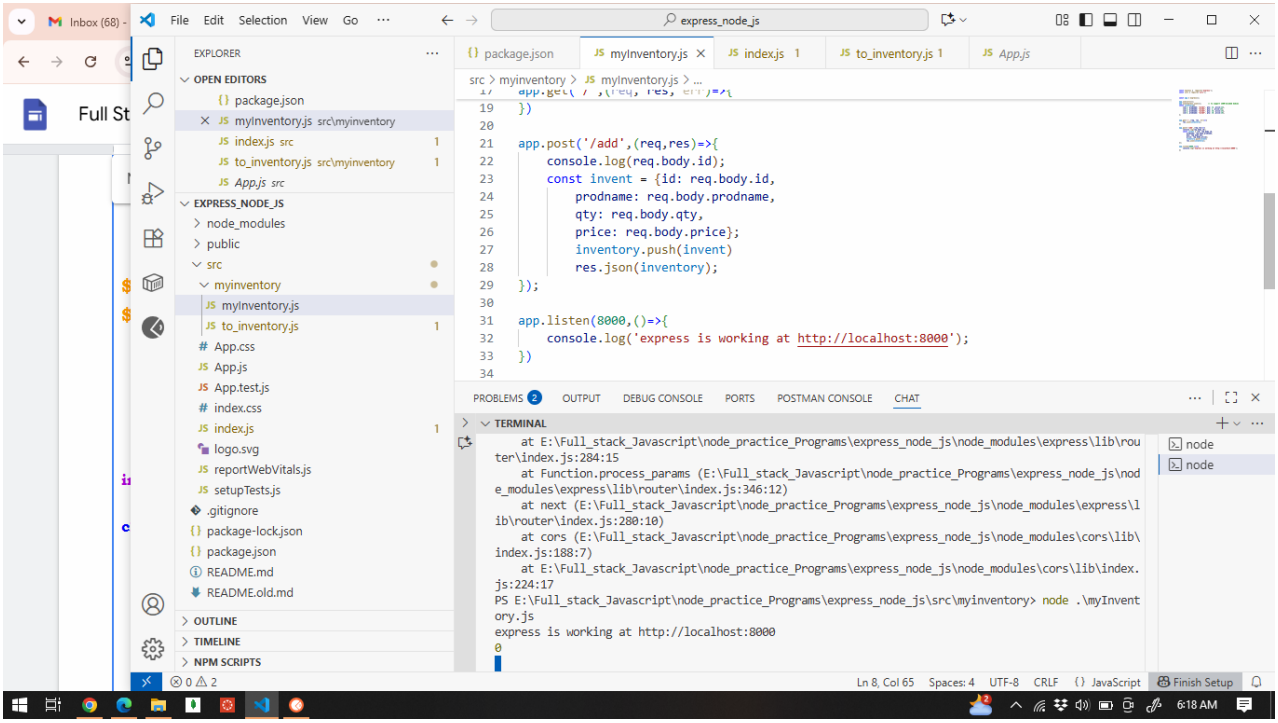
Steps for execution:

$npm start

> prog3@0.1.0 start

> react-scripts start

Expected OUTPUT



Deploy connectivity between React and Node Application for Inventory Management system

Prelims which is needed before running the programs:

$npx create-react-app program6
$npm install express cors axios

Example Program: myinventory.js

```
const express =  require('express');
const cors = require('cors');

const app = express();

app.use(cors());
app.use(express.json());       // to support JSON-encoded bodies
const inventory=[
   {id:1, prodname: "prod1", qty: 12, price:12},
   {id:2, prodname: "prod2", qty: 1, price:13},
   {id:3, prodname: "prod3", qty: 10, price:14},
   {id:4, prodname: "prod4", qty: 19, price:16},
];


app.get('/',(req, res, err)=>{
   res.json(inventory);
})

app.post('/add',(req,res)=>{
   console.log(req.body.id);
   const invent = {id: req.body.id,
      prodname: req.body.prodname,
      qty: req.body.qty,
```

```
        price: req.body.price};
        inventory.push(invent)
        res.json(inventory);
});

app.listen(8000,()=>{
   console.log('express is working at http://localhost:8000');
})
```

Steps for execution:

```
$node myinventory.js
```

Example Program: to_inventory.js

```
import React, { useState } from "react";
import axios from "axios";

function Display_Inventory() {
  const [res, setres] = useState([]);
  const resp = async () => {
    const response = await axios.get("http://localhost:8000");
    console.log("asdfasdf", response.data)
    setres(response.data);
  }
  resp();
  return (
    <div>
      <h1>Inventory Management</h1>
      <table>
        <thead>
          <tr>
            <th>ID</th>
            <th>Product Name</th>
            <th>Quantity</th>
            <th>Price</th>
          </tr>
        </thead>
        <tbody>
          {res.map((inventory) => (
            <tr key={inventory.id}>
              <td>{inventory.id}</td>
              <td>{inventory.name}</td>
              <td>{inventory.qty}</td>
              <td>{inventory.price}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}


function AddInventory() {
```

```
  const [ id, setId ] = useState(0);
  const [ name, setName ] = useState('');
  const [ qty, setQty ] = useState(0);
  const [ price, setPrice ] = useState(0);
  const [inventory, setInventory] =useState([]);
  const SubmitEvent = () => {
    const fo = { "id":id,"name": name,"qty":qty, "price":price };
    console.log("fo ",fo);
    const resp = async () => {
      const response = await axios.post("http://localhost:8000/add", fo);
      console.log(response.data)
      const getresponse = await axios.get("http://localhost:8000");
      setInventory(getresponse.data);
    }
    resp();
  }
  return (
    <div>
      <h1>Inventory Management1</h1>

        <table>
          <thead>
            <tr>
              <th>ID</th>
              <th>Product Name</th>
              <th>Quantity</th>
              <th>Price</th>
            </tr>
          </thead>
          <tbody>
            <tr>
              <td>id</td><td><input type="number"
                name="id"
                value={id}
                onChange={(e) => setId(e.target.value)}
              /></td></tr>
            <tr><td>Product Name</td><td><input type="text"
              name="prodname"
              value={name}
              onChange={(e) => setName(e.target.value)} /></td>
            </tr>
            <tr><td>Quantity</td><td><input type="number"
              name="qty"
              value={qty}
              onChange={(e) => setQty(e.target.value)} /></td>
            </tr>
            <tr><td>Price</td><td><input type="number"
              name="price"
              value={price}
              onChange={(e) => setPrice(e.target.value)} /></td></tr>
            <tr><td><input type="submit" name="onSubmit" onClick={SubmitEvent}
/></td></tr>
          </tbody>
        </table>
```

```
    </div>
  );
}

export { Display_Inventory, AddInventory };
```

index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import { AddInventory, Display_Inventory } from './to_inventory';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <Display_Inventory/>
    <AddInventory />
  </React.StrictMode>
);
```

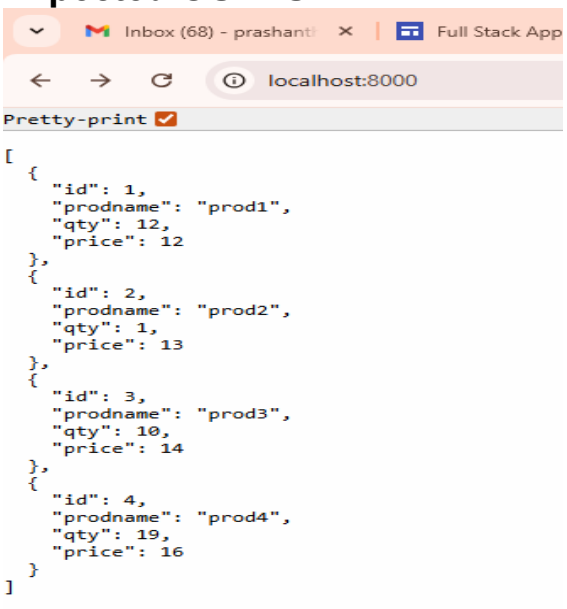
Execution Procedure:

$npm start

> prog6@0.1.0 start

> react-scripts start

Expected OUTPUT




```
[
  {
    "id": 1,
    "prodname": "prod1",
    "qty": 12,
    "price": 12
  },
  {
    "id": 2,
    "prodname": "prod2",
    "qty": 1,
    "price": 13
  },
  {
    "id": 3,
    "prodname": "prod3",
    "qty": 10,
    "price": 14
  },
  {
    "id": 4,
    "prodname": "prod4",
    "qty": 19,
    "price": 16
  }
]
```


Program 7: Develop a MongoDB query for comparison selectors, Logical Selectors for Company database

$mongosh

Setup: Sample employees Collection 🏢

First, let's insert some sample data into an employees collection to use for our queries.

```
db.employees.insertMany([
  {
    employee_id: 101,
    name: "Sonia Rao",
    department: "Engineering",
    salary: 95000,
    hire_date: new Date("2023-01-15"),
    performance_score: 4.5,
    status: "Active"
  },  {
    employee_id: 102,
    name: "Rajesh Kumar",
    department: "HR",
    salary: 68000,
    hire_date: new Date("2024-03-22"),
    performance_score: 4.1,
    status: "Active"
  },  {
    employee_id: 103,
    name: "Arjun Singh",
    department: "Engineering",
    salary: 120000,
    hire_date: new Date("2022-06-01"),
    performance_score: 4.8,
    status: "Active"
  },  {
    employee_id: 104,
    name: "Priya Mehta",
    department: "Sales",
    salary: 85000,
    hire_date: new Date("2023-11-10"),
    performance_score: 4.2,
    status: "Active"
  },  {
    employee_id: 105,
    name: "Vikram Reddy",
    department: "Sales",
    salary: 85000,
    hire_date: new Date("2024-08-01"),
    status: "On Leave"
  }
]);
```

1. Comparison Selectors

These operators compare the value of a field against a specified value.

$gt (Greater Than) & $lte (Less Than or Equal To)

- **Goal:** Find all employees with a **salary** greater than $90,000 but less than or equal to $120,000.
- **Query:**

```
db.employees.find({
  salary: { $gt: 90000, $lte: 120000 }
})
```

- **Explanation:** This query finds documents where the **salary** is between 90,001 and 120,000, inclusive. It will find Sonia Rao and Arjun Singh.

$in (In an Array)

- **Goal:** Find all employees who work in either the "HR" or "Sales" **department**.
- **Query:**

```
db.employees.find({
  department: { $in: ["HR", "Sales"] }
})
```

- **Explanation:** The **$in** operator selects documents where the **department** field's value exactly matches any value in the provided array. It will find Rajesh Kumar, Priya Mehta, and Vikram Reddy.

$ne (Not Equal)

- **Goal:** Find all employees whose **status** is not "On Leave".
- **Query:**

```
db.employees.find({
  status: { $ne: "On Leave" }
})
```

- **Explanation:** The **$ne** operator selects all documents where the **status** field value is not equal to the specified value.

2. Logical Selectors

These operators are used to combine multiple query conditions.

$and (Implicit)

- **Goal:** Find employees in the "Engineering" **department** with a **performance_score** of 4.5 or higher.
- **Query:**

```
db.employees.find({
  department: "Engineering",
  performance_score: { $gte: 4.5 }
})
```

- **Explanation:** When you list multiple fields separated by a comma in a find() query, MongoDB treats it as an **$and** condition by default. This is the most common way to combine conditions. This query will find Sonia Rao and Arjun Singh.

$or

- **Goal:** Find employees who are either in the "HR" **department** OR have a **salary** greater than $100,000.
- **Query:**

```
db.employees.find({
  $or: [
    { department: "HR" },
    { salary: { $gt: 100000 } }
  ]
})
```

- **Explanation:** The **$or** operator takes an array of condition objects and selects documents that satisfy *at least one* of the conditions. This will find Rajesh Kumar (matches department) and Arjun Singh (matches salary).

3. Combining Selectors for a Complex Query ✨

Real-world scenarios often require combining multiple operators.

- **Goal:** Find all **Active** employees in either the **Engineering** or **Sales** departments who were hired **before 2024**.
- **Query:**

```
db.employees.find({
  status: "Active",
  department: { $in: ["Engineering", "Sales"] },
  hire_date: { $lt: new Date("2024-01-01") }
})
```

Explanation: This query implicitly uses $and to combine three conditions:

1. The **status** must be "Active".
2. The **department** must be one of the values in the $in array
3. The **hire_date** must be less than ($lt) the specified date.

Program 8: Execute aggregation pipeline and its operation to illustrate text search on catalog data collection

Prelims which is needed before running the programs:

$mongosh

Step 1: Prepare Your Data Collection
First, you need a collection with text-rich fields to search. We'll use a catalog collection for this example. The key is to have fields like name and description that contain the words you want to find.
Action: In mongosh, connect to your database and run the following command to insert sample data.

```
db.catalog.insertMany([
  {
    item_id: "A001",
    name: "Classic Leather Backpack",
    description: "Durable leather backpack with multiple compartments, perfect for
daily commute or light travel.",
    tags: ["leather", "backpack", "travel", "work"],
    price: 120.00,
    in_stock: true
  },
  {
    item_id: "A002",
    name: "Ergonomic Office Chair",
    description: "Adjustable office chair with lumbar support and breathable mesh,
ideal for long working hours.",
    tags: ["office", "chair", "ergonomic", "comfort"],
    price: 280.00,
    in_stock: false
  },
  {
    item_id: "A003",
    name: "Durable Cotton T-Shirt",
    description: "A soft and durable t-shirt made from 100% organic cotton. Eco-
friendly and comfortable.",
    tags: ["cotton", "tshirt", "organic", "eco-friendly"],
    price: 25.00,
    in_stock: true
  }
]);
```

Step 2: Create a text Index
This is the most critical step for enabling text search on a Community Server. You must tell MongoDB which fields to prepare for searching. Without this index, the $text operator will not work.
Why is this needed? A text index acts like an index at the back of a book. It lists all the significant words from your specified fields and maps them back to the documents they

came from. This allows MongoDB to find words very quickly without scanning every single document in the collection.
Important Note: A collection can have **at most one** text index.
Action: Run the createIndex command. Here, we are indexing the name, description, and tags fields.

```
db.catalog.createIndex({
  name: "text",    description: "text",    tags: "text"
})
```

Step 3: Build and Execute the Aggregation Pipeline
Now, you'll construct the query. Instead of the $search stage used in Atlas, you will use the **$match** stage combined with the **$text** operator.
Action: Let's build a pipeline to find all items that contain the words "durable" or "backpack".

1. **Stage 1: The $match Stage with $text** This stage filters the documents.
    - **$text**: The operator that performs the search on the text index.
    - **$search**: The string containing the words you're looking for. By default, MongoDB searches for documents that contain *any* of the space-separated words (an OR condition).

```
{
  $match: {
    $text: {
      $search: "durable backpack"
    }
  }
}
```

Stage 2: The $project Stage This stage refines the output. It's good practice to select only the fields you need and to view the search relevance score.

- **$meta: "textScore"**: This special expression retrieves the relevance score calculated by the $text search. A higher score means a better match.

```
{
  $project: {
    _id: 0,
    name: 1,
    price: 1,
    score: { $meta: "textScore" }
  }
}
```

Complete Pipeline Execution: Combine these stages and run the aggregate command.

```
db.catalog.aggregate([
  // Stage 1: Filter documents using the text index
  {
    $match: {
      $text: {
```

```
      $search: "durable backpack"
    }
   }
  },
  // Stage 2: Reshape output and include the relevance score
  {
   $project: {
    _id: 0,
    name: 1,
    price: 1,
    score: { $meta: "textScore" }
   }
  }
])
```

Step 4: Analyze the Results

The output will be the documents that matched your search query, formatted according to your $project stage.

Expected Output:

```
[
  {
    "name": "Classic Leather Backpack",
    "price": 120,
    "score": 1.5
  },
  {
    "name": "Durable Cotton T-Shirt",
    "price": 25,
    "score": 0.75
  }
]
```

Program 9: Design an employee Management system using RESTFULL APIs in React

Prelims which is needed before running the programs:

$npx create-react-app program9
$npm install axios

db.json

```
{
  "employees": [
    { "id": 1, "firstName": "John", "lastName": "Doe", "email":
"john.doe@example.com" },
    { "id": 2, "firstName": "Jane", "lastName": "Smith", "email":
"jane.smith@example.com" }
  ]
}
```

```
$npx json-server --watch db.json --port 5000
```

Example - /src/App.js

```
import React, { useState, useEffect } from 'react';
import axios from 'axios';
import './App.css'; // For basic styling

// API URL
const API_URL = "http://localhost:5000/employees";

function App() {
    // State to hold the list of employees
    const [employees, setEmployees] = useState([]);

    // State for the form data. 'null' if we are not editing/adding.
    const [currentEmployee, setCurrentEmployee] = useState(null);

    // Fetch employees when the component loads
    useEffect(() => {
        fetchEmployees();
    }, []);

    // R: Read employees from the API
    const fetchEmployees = async () => {
        const response = await axios.get(API_URL);
        setEmployees(response.data);
    };

    // C & U: Handle form submission for Create and Update
    const handleSave = async (employee) => {
        if (employee.id) {
            // Update existing employee
            await axios.put(`${API_URL}/${employee.id}`, employee);
        } else {
            // Create new employee
            await axios.post(API_URL, employee);
        }
        fetchEmployees(); // Refresh the list
        setCurrentEmployee(null); // Hide the form
    };

    // D: Delete an employee
    const handleDelete = async (id) => {
        await axios.delete(`${API_URL}/${id}`);
```

```
        fetchEmployees(); // Refresh the list
    };

    // Handlers to show the form
    const handleAdd = () => {
        setCurrentEmployee({ firstName: '', lastName: '', email: '' }); // Show form with
empty fields
    };

    const handleEdit = (employee) => {
        setCurrentEmployee(employee); // Show form with pre-filled data
    };

    // If 'currentEmployee' is not null, show the form. Otherwise, show the list.
    if (currentEmployee) {
        return <EmployeeForm employee={currentEmployee} onSave={handleSave}
onCancel={() => setCurrentEmployee(null)} />;
    }

    return (
        <div className="container">
            <h2>Employees List</h2>
            <button onClick={handleAdd}>Add Employee</button>
            <table>
                <thead>
                    <tr>
                        <th>First Name</th>
                        <th>Last Name</th>
                        <th>Email</th>
                        <th>Actions</th>
                    </tr>
                </thead>
                <tbody>
                    {employees.map(emp => (
                        <tr key={emp.id}>
                            <td>{emp.firstName}</td>
                            <td>{emp.lastName}</td>
                            <td>{emp.email}</td>
                            <td>
                                <button onClick={() => handleEdit(emp)}>Edit</button>
                                <button onClick={() => handleDelete(emp.id)}>Delete</button>
                            </td>
                        </tr>
                    ))}
                </tbody>
            </table>
        </div>
    );
}

// A simple form component defined in the same file
const EmployeeForm = ({ employee, onSave, onCancel }) => {
    const [formData, setFormData] = useState({ ...employee });
```

```
  const handleSubmit = (e) => {
    e.preventDefault();
    onSave(formData);
  };

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData(prev => ({ ...prev, [name]: value }));
  };

  return (
    <form onSubmit={handleSubmit}>
      <h2>{formData.id ? 'Edit Employee' : 'Add Employee'}</h2>
      <input name="firstName" value={formData.firstName}
onChange={handleChange} placeholder="First Name" required />
      <input name="lastName" value={formData.lastName}
onChange={handleChange} placeholder="Last Name" required />
      <input name="email" value={formData.email} onChange={handleChange}
placeholder="Email" required />
      <button type="submit">Save</button>
      <button type="button" onClick={onCancel}>Cancel</button>
    </form>
  );
};

export default App;
```

Expected Output:

Program 10: Create a React application using react-router-dom with multiple pages (Home, About, Contact).

Prelims which is needed before running the programs:

```
$npx create-react-app program10
$npm install react-router-dom axios cors
```

Example Program: src/pages/Home.js

```
import React from 'react';

function Home() {
```

```
  return (
    <div>
      <h1>Home Page</h1>
      <p>Welcome to the homepage of our application!</p>
    </div>
  );
}

export default Home;
```

Example Program: src/pages/About.js

```
import React from 'react';

function About() {
  return (
    <div>
      <h1>About Us</h1>
      <p>This is the about page, where you can learn more about us.</p>
    </div>
  );
}

export default About;
```

Example Program: src/pages/Contact.js

```
import React from 'react';

function Contact() {
  return (
    <div>
      <h1>Contact Page</h1>
      <p>Get in touch with us through this contact page.</p>
    </div>
  );
}

export default Contact;
```

Example Program: src/App.js

```
import React from 'react';
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';
import Home from './pages/Home';
import About from './pages/About';
import Contact from './pages/Contact';
import './App.css';

function App() {
  return (
    <BrowserRouter>
      <div className="App">
        <nav>
          <ul>
```

```
            <li>
              <Link to="/">Home</Link>
            </li>
            <li>
              <Link to="/about">About</Link>
            </li>
            <li>
              <Link to="/contact">Contact</Link>
            </li>
          </ul>
        </nav>

        <main>
          <Routes>
            <Route path="/" element={<Home />} />
            <Route path="/about" element={<About />} />
            <Route path="/contact" element={<Contact />} />
          </Routes>
        </main>
      </div>
    </BrowserRouter>
  );
}

export default App;
```

Execution Procedure:

$npm start

> prog6@0.1.0 start

> react-scripts start

Expected OUTPUT